

LCC RPiPico Node (v3.0)

An open, DIY-friendly Raspberry Pi Pico platform for OpenLCB / LCC model railroad control

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The LCC RPiPico Node is a single, general-purpose board design that replaces a whole family of earlier dedicated boards. Specialized functions — displays, stepper drivers, servos, and more — plug in as breakout boards rather than requiring a new board revision for every node type. It grew out of LocoNet turntable controllers dating back to 2021 and migrated to full LCC/OpenLCB support in 2024.

WHY IT'S AN EASY DIY ENTRY POINT

- **Arduino IDE** — programmed with the standard Earle Philhower RP2040 board package; no CMake/toolchain setup.
- **Full LCC/OpenLCB** stack included via the OpenLcbCLib library — alias allocation, datagrams, CDI, memory config, and more already implemented.
- **One board**, every node type — swap breakout boards instead of re-designing hardware for each new function.
- **Hand-solderable** — discrete MCP2518/MCP2562 CAN chips instead of hard-to-hand-solder integrated parts.
- **Documented pin/architecture standard** shared across every example project, so patterns learned on one node transfer directly to the next.
- **Working example firmware to copy** — four complete, real-world node applications with full source (see right).

EXAMPLE NODE PROJECTS (OPEN SOURCE, FULL FIRMWARE ON GITHUB)

- **Turntable** — Motorized turntable controller — stepper-driven positioning, home/limit sensing, touchscreen UI with track selection, bridge animation, and RailCom occupancy display.
- **Roundhouse** — Multi-stall roundhouse door control — servo-driven doors synchronized to turntable track selection.
- **PixelLights** — Addressable NeoPixel lighting controller — scene/effect control for structure, signal, or ambient layout lighting over LCC events.
- **Clock Lights** — Layout lighting tied to the LCC fast clock — day/night lighting transitions and dimming driven by the shared broadcast fast-clock protocol.

GET STARTED

- **Board hardware** (schematics, PCB, gerbers, breakout designs): github.com/bocabob/LCC-RPi-Pico-Board
- **Firmware architecture standard & example applications**: github.com/bocabob/LCC_RPiPico_Common
- **Board support**: Earle Philhower's RP2040 core (arduino-pico) — install via the Arduino IDE Boards Manager
- **LCC protocol stack**: OpenLcbCLib (Jim Kueneman) / OpenLCB Single Thread — linked from the Common repo above

KEY BOARD ATTRIBUTES

- **Processor**: Raspberry Pi Pico or Pico 2 (Pico 2 recommended)
- **CAN**: discrete MCP2518 controller + MCP2562 transceiver
- **Power**: dual 3.3V/5V switching regulation; power from LCC bus or 8–15V external; can inject power onto the bus
- **NVM**: I2C EEPROM or FRAM (onboard footprint or plug-in module) — spares the Pico's flash from wear
- **I/O**: 16 GP pins via two 10-pin IDC connectors (I/O-1, I/O-2) + a 6-pin analog header (I/O-3)
- **Breakout ecosystem**: touchscreen displays (SPI or parallel), stepper driver, servo, and test/ribbon breakouts
- **Buttons**: onboard reset (Pico has none) + header for a remote reset button; Blue/Gold user buttons where pins allow

